Testing and Debugging

Purpose and scope

We ran 73 test cases on the Passenger Van Rental System. 70 passed and 3 failed. The tests check that each menu option (1 to 16) works, that wrong input is rejected with a clear message, and that saved data loads again when the program is reopened. The failed tests are explained in the debugging records.

Test environment and method

- **Program:** the CSC104-RENTALPROJECT source code, compiled with (OpenJDK) and started with java -cp bin Main
- **Method:** every test was done on the running program by typing the inputs in the menu. A script typed the inputs so each test could be repeated the same way.
- **Starting data:** each test began with a fresh copy of the data files: 25 vans, 15 customers, 3 rental staff (E1, E2, E3), no rentals and no maintenance records.
- **Changed data:** a few tests need an old rental or a bad file. These are marked "Setup" in the Test case column.
- **Test date:** 2026-10-26. Rental dates come from the computer date, so dates in the results follow this day.
- **Status:** Pass means the actual result matched the expected result. Fail means it did not.

Test summary

Group	Test Cases	Passed	Failed
Customer (TC-01 to TC-18)	18	18	0
Van (TC-19 to TC-29)	11	11	0
Rental (TC-30 to TC-43)	14	13	1
Return (TC-44 to TC-49)	6	6	0
Overdue and SMS (TC-50 to TC-54)	5	5	0
Maintenance (TC-55 to TC-64)	10	10	0
Menu (TC-65 to TC-67)	3	3	0
File persistence (TC-68 to TC-73)	6	4	2
Total	**73**	**70**	**3**

The three failed tests are TC-43 (no limit on rental days), TC-71 (crash on a blank line in vans.txt) and TC-73 (rental history lost when a van is deleted).

Test cases

The tables below list every test with its input, the expected result, the actual result and the status.

Customer tests

ID	Test case and input	Expected result	Actual result	Status
TC - 01	Add a customer with valid details. Input: Anna Lim, 09181234567, anna.lim@gmail.com, Purok 5; Libertad; Butuan City, Driver's License, N04-12-345678	Message "Customer added. ID: C16". Customer is saved in customers.txt	Message "Customer added. ID: C16" shown. Line for C16 found in customers.txt	Pass
TC - 02	Add a customer with an empty name. Input: (blank), then Anna Lim	Empty name is rejected and the system asks again	Message "Input cannot be empty or contain commas." shown. Anna Lim was then accepted as C16	Pass
TC - 03	Add a customer with a comma in the name. Input: Lim, Anna	Comma is rejected because it would break the data file	Same message shown and the system asked again	Pass
TC - 04	Contact number with letters. Input: 09abcdefghi	Message "Invalid number. Use 11 digits starting with 09."	Message shown. A valid number was accepted next	Pass
TC - 05	Contact number with 10 digits. Input: 0918123456	Number is rejected	Message "Invalid number. Use 11 digits starting with 09." shown	Pass
TC - 06	Email without the @ sign. Input: annalim.gmail.com	Message "Invalid email format."	Message shown and the system asked again	Pass
TC - 07	Address that is too short. Input: Butuan	Message "Please enter the complete address."	Message shown and the system asked again	Pass
TC - 08	ID type outside the list. Input: 9	Message "Please choose a number from the list."	Message shown. Choice 1 was accepted next	Pass

TC - 09	ID type typed as letters. Input: abc	Message "Please enter a whole number." and no crash	Message shown and the system asked again	Pass
TC - 10	ID number that is too short. Input: 12	Message "Invalid ID number. Use 5 to 20 letters, digits, or dashes."	Message shown and the system asked again	Pass
TC - 11	View all customers (option 8)	15 customers listed. ID numbers are hidden except the last 4 characters	C1 to C15 listed. ID numbers shown like ****2222	Pass
TC - 12	Update customer C1 with new valid details. Input: 09999999999, kien.new@gmail.com, new address, Passport, P7654321A	Message "Customer updated." New details appear in the list and in customers.txt	Message shown. customers.txt has C1 with 09999999999 and kien.new@gmail.com	Pass
TC - 13	Update a customer ID that does not exist. Input: C99	Message "Error: Customer ID not found."	Message shown	Pass
TC - 14	Delete a customer with no rental. Input: C15, then Y	Message "Customer deleted." C15 is gone from the list	Message shown. C15 not in the list	Pass
TC - 15	Delete a customer but answer no. Input: C15, then N	Message "Delete cancelled." C15 stays in the list	Message shown. C15 still in the list	Pass
TC - 16	Delete a customer ID that does not exist. Input: C99	Message "Error: Customer ID not found."	Message shown	Pass
TC - 17	Delete a customer who has a rental. Input: rent and return a van for C1 (R1), then delete C1	Delete is refused with "Cannot delete a customer that has rental records (R1)."	Message shown. Customer was not deleted	Pass
TC - 18	Reuse of a customer ID. Input: delete C3, then add a new customer	New customer gets the free ID C3 and the list stays in order	Message "Customer added. ID: C3". List shows C1, C2, C3 in order	Pass

Van tests

ID	Test case and input	Expected result	Actual result	Status
TC - 19	Add a Standard van. Input: S, new 111, Toyota Hiace, 15	Message "Van added." Plate is saved in capital letters (NEW-111)	Message shown. vans.txt has the line Standard,NEW-111,Toyota Hiace,15,true,false	Pass
TC - 20	Add a Luxury van. Input: L, LUX-999, Hyundai Staria, 9	Message "Van added." Van is saved as Luxury	Message shown. vans.txt has the line Luxury,LUX 999,Hyundai Staria,9,true,false	Pass
TC - 21	Add a van with a plate that already exists. Input: abc 123	Message "Error: A van with that plate number already exists." Small and capital letters count the same	Message shown. Van was not added	Pass
TC - 22	Add a van with a wrong type letter. Input: X	Message "Error: Type must be S or L."	Message shown	Pass
TC - 23	Add a van with 0 seats. Input: 0	Message "Error: Seats must be at least 1." Van is not added	Message shown	Pass
TC - 24	Seats typed as letters. Input: abc, then 12	Message "Please enter a whole number." then 12 is accepted	Message shown, then "Van added."	Pass
TC - 25	Delete an available van, then search for it. Input: EFG 208	Message "Van deleted." Search then says "Van not found."	Both messages shown	Pass
TC - 26	Delete a plate that does not exist. Input: ZZZ-000	Message "Error: Van not found."	Message shown	Pass
TC - 27	Search a van using small letters. Input: abc-123	Van details and "Status: Available - can be rented."	Details of ABC-123 shown with that status	Pass
TC - 28	Search a plate that does not exist. Input: ZZZ-000	Message "Van not found."	Message shown	Pass
TC - 29	View all vans (option 6)	Three groups: AVAILABLE (25), RENTED (0), UNDER MAINTENANCE (0)	The three groups were shown with those counts	Pass

Rental tests

ID	Test case and input	Expected result	Actual result	Status
TC - 30	Rent a Standard van for 3 days. Input: E1, C1, ABC-123, 3	Rental R1 is created. Fee is P3000 x 3 = P9000.0. Email receipt is shown and saved in notifications_log.txt	"Rental created: R1" shown. Receipt shows Total fee P9000.0. Receipt and rentals.txt line were saved	Pass
TC - 31	Rent a Luxury van for 2 days. Input: E2, C2, GHI-789, 2	Fee is P4500 x 2 + P500 cleaning fee = P9500.0	Receipt shows Total fee P9500.0	Pass
TC - 32	Rent with a staff ID that does not exist. Input: E9	Message "Error: Rental staff ID not found. Valid IDs: E1, E2, E3."	Message shown	Pass
TC - 33	Rent with a customer ID that does not exist. Input: E1, C99	Message "Error: Customer ID not found."	Message shown	Pass
TC - 34	Rent a plate that does not exist. Input: E1, C1, ZZZ-000	Message "Error: Van not found."	Message shown	Pass
TC - 35	Rent a van that is already rented. Input: rent ABC-123 to C1, then try ABC-123 for C2	Second rent is refused with "NOT AVAILABLE: it is already rented."	Message shown. ABC-123 was not in the available list	Pass
TC - 36	Rent for 0 days. Input: 0	Message "Error: Rental days must be at least 1." No rental is saved	Message shown. rentals.txt stayed empty	Pass
TC - 37	Rent for a negative number of days. Input: -5	Message "Error: Rental days must be at least 1."	Message shown	Pass
TC - 38	Days typed as letters. Input: three, then 3	Message "Please enter a whole number." then 3 is accepted	Message shown, then "Rental created: R1"	Pass
TC - 39	Rent to a customer with no address or ID on file. Setup: customer C16 in the old 4-field format	Rent is refused with "This customer has no address or valid ID on file."	Message shown. Old file format was loaded without error	Pass
TC - 40	Rent a van that is under maintenance. Input: send ABC-123 to maintenance, then rent it	Rent is refused with "NOT AVAILABLE: it is under maintenance."	Message shown	Pass

TC - 41	Van status after renting. Input: rent ABC-123, then option 6	ABC-123 moves to RENTED. Counts: AVAILABLE (24), RENTED (1)	Counts shown as expected	Pass
TC - 42	View rentals when there are none (option 11)	Message "No rentals."	Message shown	Pass
TC - 43	Rent a van for a very large number of days. Input: 2147483647	The system should refuse the number and keep the van available	Rental was accepted. Receipt shows return date +5881637-05-05 and total fee P6.442450941E12. Van became Rented	Fail

Return tests

ID	Test case and input	Expected result	Actual result	Status
TC - 44	Return a van on time. Input: rent ABC-123 for 2 days, then return R1 with E2	"Van returned. Thank you!" and "Returned on time. No late fee."	Both messages shown	Pass
TC - 45	Return a van 7 days late. Setup: rental R1 started 10 days ago for 3 days	Late fee is P250 x 7 = P1750.0. Total to pay is P9000 + P1750 = P10750.0	"Returned 7 day(s) late." Late fee P1750.0. Total to pay P10750.0	Pass
TC - 46	Return a rental ID that does not exist. Input: R99	Message "Error: Rental ID not found."	Message shown	Pass
TC - 47	Return the same rental twice. Input: R1, then R1 again	Second return is refused with "This van was already returned."	Message shown	Pass
TC - 48	Return with a staff ID that does not exist. Input: E9	Message "Error: Rental staff ID not found."	Message shown	Pass
TC - 49	Van status after return. Input: rent and return ABC-123, then search it	"Status: Available - can be rented."	Status shown as Available	Pass

Overdue and SMS tests

ID	Test case and input	Expected result	Actual result	Status
TC - 50	Overdue list with one overdue rental (option 13). Setup: R1 started 10 days ago for 3 days	Rental is listed with customer name, contact, address, valid ID and "late fee so far P1750.0"	All details shown. Line read "Late : 7 day(s), late fee so far P1750.0"	Pass
TC - 51	Overdue list when nothing is due	Message "No rentals are due today or overdue."	Message shown	Pass
TC - 52	Send expiration SMS for an overdue rental (option 12). Setup: same overdue rental	SMS is shown and saved in the log. "1 expiration message(s) sent."	SMS to 09393818714 shown and saved in notifications_log.txt. Count message shown	Pass
TC - 53	Send expiration SMS when nothing is due	Message "0 expiration message(s) sent."	Message shown	Pass
TC - 54	Rental that ends today. Setup: R1 started 3 days ago for 3 days	Rental is listed as due but has no late line (0 days late)	Rental listed with customer details. No "Late" line shown	Pass

Maintenance tests

ID	Test case and input	Expected result	Actual result	Status
TC - 55	Send an available van to maintenance. Input: E3, GHI-789, Oil change	"Record: M1". Van status is Under maintenance and it cannot be rented. Record is in the history	Message "Record: M1" shown. Search showed "Status: Under maintenance - cannot be rented." History showed M1 by Camille Perez. maintenance.txt saved	Pass
TC - 56	Send a rented van to maintenance. Input: rent ABC-123, then maintenance for ABC-123	Refused with "That van is currently rented. Wait until it is returned."	Message shown	Pass

TC - 57	Send a van that is already under maintenance. Input: same van twice	Second try is refused with "That van is already under maintenance."	Message shown	Pass
TC - 58	Maintenance for a plate that does not exist. Input: ZZZ-000	Message "Error: Van not found."	Message shown	Pass
TC - 59	Maintenance with a staff ID that does not exist. Input: E9	Message "Error: Rental staff ID not found."	Message shown	Pass
TC - 60	Finish maintenance. Input: start and finish maintenance for ABC-123, then search it and view history	"Maintenance finished. The van can be rented again." Status is Available. Record shows Done with a date	All three results were shown	Pass
TC - 61	Finish maintenance on a van that is not in maintenance. Input: ABC-123	Message "Error: That van is not under maintenance."	Message shown	Pass
TC - 62	Delete a van that is under maintenance	Refused with "Cannot delete a van that is under maintenance."	Message shown	Pass
TC - 63	Delete a van that is rented	Refused with "Cannot delete a van that is currently rented."	Message shown	Pass
TC - 64	View maintenance history when empty (option 16)	Message "No maintenance records."	Message shown	Pass

Menu tests

ID	Test case and input	Expected result	Actual result	Status
TC - 65	Menu choice not on the list. Input: 99	Message "Invalid choice. Try again." and the menu shows again	Message shown and menu came back	Pass
TC - 66	Menu choice typed as letters. Input: abc	Message "Please enter a whole number." and the program keeps running	Message shown	Pass

TC - 67	Exit the program. Input: 0	Message "Saving data... Goodbye!"	Message shown	Pass

File persistence tests

ID	Test case and input	Expected result	Actual result	Status
TC - 68	Data after closing and opening the program. Run 1: add a customer, rent ABC-123, send GHI-789 to maintenance, exit. Run 2: view customers, rentals, maintenance and vans	All saved data is loaded again in run 2	Run 2 showed customer C16, rental R1 (Active), maintenance record M1, ABC-123 as Rented and GHI-789 as Under maintenance	Pass
TC - 69	Start the program when the data files are missing. Setup: delete all five data files	Program does not crash. It loads the 3 default vans and the default staff, and shows no customers	AVAILABLE (3) vans shown and "No customers." No crash	Pass
TC - 70	A rented van stays rented after restart. Run 1: rent ABC-123, exit. Run 2: try to rent ABC-123 to C2	Van is still in RENTED and cannot be rented again	RENTED (1) shown and "NOT AVAILABLE: it is already rented."	Pass
TC - 71	Blank line at the end of vans.txt. Setup: add one empty line to vans.txt	Program starts and ignores the blank line	Program crashed at start-up with ArrayIndexOutOfBoundsException in FileManager.loadVans (line 60). No menu was shown	Fail
TC - 72	Wrong number in vans.txt. Setup: add the line Standard,BAD-001,Test Van,abc,true,false	Message "Error loading vans" and the program keeps running	Message "Error loading vans: For input string: "abc"" shown. Program kept running with the 25 good vans	Pass

TC - 73	Delete a van that has a returned rental, then restart. Run 1: rent ABC-123, return it, delete ABC-123, exit. Run 2: view rentals	Rental history stays complete (R1 still listed)	Van was deleted with no warning. In run 2 the rentals list said "No rentals." The history of R1 was lost	Fail

Debugging records

Finding and root cause	Proposed correction	Retest Plan
BUG-01 (TC-71): FileManager.loadVans reads p[1] on a blank line and throws ArrayIndexOutOfBoundsException. The method only catches IOException and NumberFormatException, so the program crashes at start-up and no menu is shown	Skip blank lines and lines with fewer than 5 fields. Also catch RuntimeException, like the other load methods	Repeat TC-71. Also test a short line and a normal file, and check that the program starts and shows 25 vans. OPEN
BUG-02 (TC-73): RentalSystem.deleteVan only checks if the van is rented or under maintenance. FileManager.loadRentals skips any rental whose van is missing, so the rental history is lost on the next save	In deleteVan, refuse the delete if any rental uses the van, the same way deleteCustomer works	Repeat TC-73. Also check that deleting a van with no rental (TC-25) still works. OPEN
BUG-03 (TC-43): RentalSystem.rentVan only checks that days is 1 or more, so 2147483647 days is accepted. The receipt shows a return date in the year 5881637 and a fee of P6.4 trillion	Add a limit of 30 days. If days is more than 30, throw RentalException("Rental days cannot be more than 30.")	Repeat TC-43. Also test 1, 30 and 31 days. OPEN

Debugging status

We found the cause of each of the three failed tests by reading the code and running the test again. The fixes in the table above are our plan and have not been added to the program yet. Next, we will change FileManager.java and RentalSystem.java, run the same tests again, and write the new results in this document.

Coverage limitations and evidence

The test set covers customer, van, rental, return and maintenance operations, input checks, not-found and duplicate errors, van status changes, late fees, different fees for Standard and Luxury vans, email and SMS messages, saving and loading of data, and basic recovery from wrong input in the console menu.

Reproducing the tests

The file TestRental.java runs all 73 test cases on the real program. It starts Main for each test, types the menu inputs, and checks the screen output and the saved files. Each test starts with a fresh copy of the data files.

1. Put TestRental.java in the project folder (the folder that has src).
2. Compile the application sources into a classes folder: javac -d classes src/*.java
3. Compile the harness: javac -cp classes -d classes TestRental.java
4. Make an empty test-run folder and start the harness from it: mkdir testrun, then cd testrun, then java -cp ../classes TestRental
5. Compare the RESULT lines on the screen with the results.log below. The three cases TC-43, TC-71 and TC-73 should fail on the supplied baseline.

Conclusion

70 of 73 tests passed (96%). The main functions work which is customers, vans, renting, returning, late fees, maintenance, email and SMS messages, and saving and loading data. Wrong input is handled with clear messages in all tests except the three failed ones. The tests also show the OOP parts working. Standard and Luxury vans give different fees through the same method (TC-30 and TC-31). The custom RentalException gives clear errors in most tests. Data is saved to and loaded from the text files (TC-68 and TC-70). We recommend fixing BUG-02 first, because it can lose rental history, then BUG-01 and BUG-03.